

Dispersion in Financing Cost and Development

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This paper

Motivations:

- Most aggregate theories of financial frictions model credit available at a single cost of financing but rationed.
- High levels and high dispersion in spreads, especially in developing countries.

Question: What's the role of level and dispersion of financing spreads on economic development and firm dynamics?

- Empirically, using Brazilian data to investigate sources contributing to the dispersion of spreads.
- Quantitatively, introducing financing spreads into an occupational choice model of credit constrained entrepreneurs.

Contributions (key features):

- spreads for universal of loans to formal firms for a developing economy
- dispersion in interest rate + self-financing
- first to link financial details to firm dynamics and use both spreads and firm growth to evaluate the mechanism

Data:

- ① Brazilian Public Credit Registration:
 - ▶ info on bank loans to all formal firms from January 2005 - December 2016
 - ▶ firm level spread = credit-weighted average of firm's outstanding loans rates
 - one-year interbank deposit rate
- ② RAIS: a matched employer-employee administrative dataset covering all formal firms in Brazil.

Empirics: High level and dispersion of spreads (loan level)

- Here, spread = loan interest rate - one-year interbank deposit rate

Table A1: Summary Statistics

A-3

	Obs	Mean	Median	St. Dev.	Min.	P5	P95	Max.
Outst. loan	11,846,248	909,708	60,414	6.92e+07	0.001	3,781	1,093,925	1.12e+11
Spread	11,846,248	44.43	23.47	60.41	-10.23	-3.05	176.74	321.28
Maturity	11,846,248	1.20	0.66	2.05	0	0.08	3.71	7.32
NPL	11,846,248	0.043	0.000	0.122	0	0	0.253	1
Currency dum.	11,846,248	0.037	0	0.190	0	0	0	1
Firm-bank rel.	11,846,248	80.16	45.55	120.67	0	1.49	246.98	1,407
N. of banks	11,846,248	2.12	2	1.415	1	1	5	36
Size	11,846,248	21.03	4	279.10	0	0	49	105,455
Age	11,846,236	14.48	9.24	19.29	0	1.52	38.50	118.62
Wage bill	11,846,248	33,798	3,505	989,213	0	0	61,411	1.10e+09
Got loan	11,846,248	0.659	1	0	0	1	1	-0.67
Firm growth	9,167,466	0.010	0	0.534	-11.20	-0.693	0.754	8.18
Exit	11,284,288	0.082	0	0.274	0	0	1	1

Empirical Analysis

Q: How much the variation in the (firm level) spreads can be explained by loan characteristics and observed risk?

Table 1: Spreads† and Loan Characteristics..

	(1)	(2)	(3)	(4)
Constant	0.4285*** [41.77]	0.2583*** [23.15]	0.2153*** [27.45]	0.1255*** [13.21]
→ Maturity	-0.0025*** [-27.50]	-0.0004*** [-17.57]	-0.0004*** [-15.70]	-0.0004*** [-16.03]
→ NPL	0.2973*** [40.52]	0.1604*** [20.11]	0.1372*** [15.83]	0.1367*** [15.55]
NPL lag (mean)	0.1259*** [13.95]	0.0639*** [9.93]	0.0789*** [13.82]	0.0758*** [13.63]
NPL lead (mean)	0.0894*** [12.20]	0.0228*** [4.13]	0.0491*** [10.57]	0.0488*** [10.64]
Currency dummy	-0.0096** [-2.59]	-0.0061* [-1.72]	0.0026 [1.15]	0.0025 [1.07]
→ Firm-bank Relationship	-0.0001*** [-25.86]	-0.0001*** [-30.44]	-0.0001*** [-29.91]	-0.0001*** [-31.99]
→ Number of banks	-0.0433*** [-27.25]	-0.0204*** [-18.25]	-0.0162*** [-25.23]	-0.0145*** [-21.22]
Observations	11,846,248	11,846,248	11,846,248	11,846,248
R-squared	0.121	0.454	0.487	0.502
Adj R-squared	0.121	0.454	0.487	0.495
Loan type controls	No	Yes	Yes	Yes
Sector FE	No	No	Yes	Yes
State FE	No	No	Yes	Yes
Firm-type FE	No	No	Yes	Yes
Time FE	No	No	Yes	Yes
Sector x State x Time FE	No	No	No	Yes
Earmarked dummy	No	No	No	Yes
RMSE	0.301	0.237	0.230	0.228

† The dependent variable is the $\log(1 + \text{spread})$, where spreads are calculated at the firm level as the credit-weighted average spread on outstanding loans.

t statistics in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Empirical Analysis

Q: How much of the remaining variation can be explained by firm characteristics (e.g. age and size)?

Table 2: Spreads Residuals† and Firm characteristics

	(1)	(2)	(3)	(4)	(5)
Constant	-0.0000*** [-586,352.92]	0.0067*** [57.04]	0.0416*** [12.62]	0.0949*** [232.22]	0.0307*** [5.37]
Age		-0.0007*** [-66.92]	-0.0041*** [-12.31]	-0.0000*** [-2.94]	-0.0042*** [-13.24]
Age ²		0.0000*** [57.59]	0.0000*** [11.63]	0.0000*** [4.30]	0.0000*** [12.50]
Wagebill (ln)				-0.0113*** [-229.28]	0.0015*** [2.75]
Observations	11,846,248	11,846,236	11,846,236	10,850,982	10,850,982
Firm FE	Yes	No	Yes	No	Yes
R-squared	0.538	0.000	0.539	0.005	0.541
Adj. R-squared	0.375	0.000417	0.377	0.00543	0.376
RMSE	0.179	0.226	0.179	0.225	0.178

† Spread residuals are those from Model (4) in Table 1

t statistics in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Model: Environment

Endowments: $(a(t), z(t))$

- make an occupational choice: worker or entrepreneur
- $\mu(z) = \eta z^{-(\eta+1)}$, $z \geq 1$; with Poisson arrival rate γ , draw new z from μ

Preferences:

- $E_0 \left[\int_0^\infty e^{-\rho t} u(c(t)) dt \right]$
- $u(c(t)) = \frac{c(t)^{1-\sigma} - 1}{1-\sigma}$, $\sigma > 0$

Technology:

- $y(t) = z(t)k(t)^\alpha n(t)^\theta$
- $\theta, \alpha > 0, \theta + \alpha < 1$
- fixed cost : κ

Three sources of frictions from financial Intermediation:

- real intermediation cost:
 - ▶ $g(l, \tau(a, z)) = l\tau(a, z)$
 - ▶ $l = k - a$ and $\tau(a, z) = \tau_0 + \frac{\tau_a}{1+a} + \frac{\tau_z}{1+z}$, with $\tau_0, \tau_a, \tau_z \geq 0$
- lender market power (χ): stems from the bilateral nature of borrowing/lending opportunities. The lender and borrower negotiate over the interest rate $\tilde{r}$ and loan size l via Nash bargaining.
- limited enforcement: put constraint on the possible $(\tilde{r}, l)$ in the bargaining problem. Borrowers have the option to strategically default and lenders can only recover a fraction ϕ of the output produced net of labor cost $\Rightarrow$ quantity restriction

Model: Static Optimization

Entrepreneurial Profits: $\pi(k, n, \tilde{r}; a, z) = zk^\alpha n^\theta - wn - \tilde{r}(k - a) - ra - \kappa$

- first solve for unconstrained capital level $k^u(z)$
- if $k^u(z) > a$, solve for set of capital levels satisfying the limited enforcement constraint:

$$\underbrace{\pi(k, \tilde{r}; a, z)}_{\text{income from repayment}} \geq \underbrace{(1 - \phi)(x(k, \tilde{r}; a, z) + \tilde{r}(k - a)) - ra - \kappa}_{\text{income from defaulting}}$$

- $\pi(k, \tilde{r}; a, z) = \max_{n \geq 0} \pi(k, n, \tilde{r}; a, z)$
- $x(k, \tilde{r}; a, z) = \max_{n \geq 0} \pi(k, n, \tilde{r}; a, z) + \tilde{r}(k - a) + ra + \kappa$ which is output net of labor costs

▶ hard quantity constraint: $\bar{k}(\tilde{r}; a, z)$

- Bargaining problem:

$$\bar{k} \geq k \geq a, \tilde{r} \quad \underbrace{[(\tilde{r} - r - \tau(a, z))(k - a)]^\chi}_{S^b} \underbrace{[\pi(k, \tilde{r}; a, z) + \tilde{r}(k - a) - \tilde{w}(a, z)]^{1-\chi}}_{S^e}$$

$$\tilde{w}(a, z) = \max\{w, x(a, z) - ra - \kappa\}$$

$$\Rightarrow (\tilde{r}, k^b)(a, z; \chi) \bullet$$

Model: Static Optimization

When $k^b(a, z) < \bar{k}(\tilde{r}; a, z)$,

$$k^b(a, z) = \left(z \left(\frac{\alpha}{r + \tau(a, z)} \right)^{(1-\theta)} \left(\frac{\theta}{w} \right)^\theta \right)^{\frac{1}{1-\alpha-\theta}}$$
$$\tilde{r}(a, z) = r + \tau(a, z) + \chi \left(\frac{\pi^b(a, z) - \tilde{w}(a, z)}{k^b(a, z) - a} \right)$$

- Intermediary market power will not statically distort capital but will impact the profits of the entrepreneur, and will therefore dynamically impact the entrepreneur's ability to self finance.

When $k^b(a, z) = \bar{k}(\tilde{r}; a, z)$,

$$\phi x(k, z) = \tilde{r}(k - a)$$

$$0 = \chi S^e \left(\phi \frac{\partial x(k, z)}{\partial k} - r - \tau(a, z) \right) + (1 - \chi) S^b (1 - \phi) \frac{\partial x(k, z)}{\partial k}$$

Model: Static Optimization

Proposition 1 Consider an agent (a, z) such that $a < k^u(z)$, $k < \bar{k}(\bar{r})$ and $S^e(a, z) \geq 0$.

1. Case of $\chi = 0$:

$$\tilde{r}(a, z) = r + \tau(a, z),$$

$$\frac{\partial \tilde{r}(a, z)}{\partial z} \leq 0 \text{ and } \frac{\partial k^b(a, z)}{\partial z} \geq 0; \frac{\partial \tilde{r}(a, z)}{\partial a} \leq 0 \text{ and } \frac{\partial k^b(a, z)}{\partial a} \geq 0.$$

2. Case of $\chi \in (0, 1)$ and $\tau(a, z) = \tau_0 \geq 0$: $\frac{\partial \tilde{r}(a, z)}{\partial \chi} > 0$ and $\frac{\partial k^b(a, z)}{\partial \chi} = 0$.

3. Case of $\chi = 1$ and $\tau(a, z) = \tau_0 \geq 0$:

$$\tilde{r}(a, z) = r + \tau_0 + \frac{1}{k^b(a, z) - a} (y^b(a, z) + \tau_0 a - \kappa - \tilde{w}(a, z)),$$

$$\frac{\partial \tilde{r}(a, z)}{\partial z} > 0 \text{ and } \frac{\partial k^b(a, z)}{\partial z} > 0. \text{ In addition, if } w > \tilde{\pi}(a, z), \text{ then } \frac{\partial \tilde{r}(a, z)}{\partial a} > 0 \text{ and } \frac{\partial k^b(a, z)}{\partial a} = 0; \text{ and if } w < \tilde{\pi}(a, z), \text{ then } \frac{\partial \tilde{r}(a, z)}{\partial a} < 0 \text{ and } \frac{\partial k^b(a, z)}{\partial a} = 0.$$

Proposition 2 Consider an agent (a, z) such that $a < k^u(a, z)$, the incentive compatible constraint binds and $S^e(a, z) \geq 0$.

1. Case of $\chi = 0$:

$$\tilde{r}(a, z) = r + \tau(a, z),$$

$$\frac{\partial \tilde{r}(a, z)}{\partial z} \leq 0 \text{ and } \frac{\partial k^b(a, z)}{\partial z} > 0; \frac{\partial \tilde{r}(a, z)}{\partial a} \leq 0 \text{ and } \frac{\partial k^b(a, z)}{\partial a} > 0.$$

2. Case of $\chi \in (0, 1)$ and $\tau(a, z) = \tau_0 \geq 0$: $\frac{\partial \tilde{r}(a, z)}{\partial \chi} > 0$ and $\frac{\partial k^b(a, z)}{\partial \chi} < 0$.

3. Case of $\chi = 1$ and $\tau(a, z) = \tau_0 \geq 0$:

$$\tilde{r}(a, z) = \frac{\phi x(k^b(a, z))}{k^b(a, z) - a}.$$

$$\text{If } w > \tilde{\pi}(a, z), \text{ then } \frac{\partial \tilde{r}(a, z)}{\partial z} > 0 \text{ and } \frac{\partial k^b(a, z)}{\partial z} < 0; \text{ and } \frac{\partial \tilde{r}(a, z)}{\partial a} \geq 0 \text{ and } \frac{\partial k^b(a, z)}{\partial a} > 0. \\ \text{If } w < \tilde{\pi}(a, z), \text{ then } \frac{\partial \tilde{r}(a, z)}{\partial z} \geq 0 \text{ and } \frac{\partial k^b(a, z)}{\partial z} \geq 0; \text{ and } \frac{\partial \tilde{r}(a, z)}{\partial a} \leq 0 \text{ and } \frac{\partial k^b(a, z)}{\partial a} > 0.$$

Model: Static Optimization

Instantaneous entrepreneurial profits:

$$\pi^*(a, z) = \begin{cases} x(a, z) - ra - \kappa, & \text{if } l = 0 \\ x(k^b, z) - \tilde{r}(k^b - a) - ra - \kappa, & \text{if } l > 0 \end{cases}$$

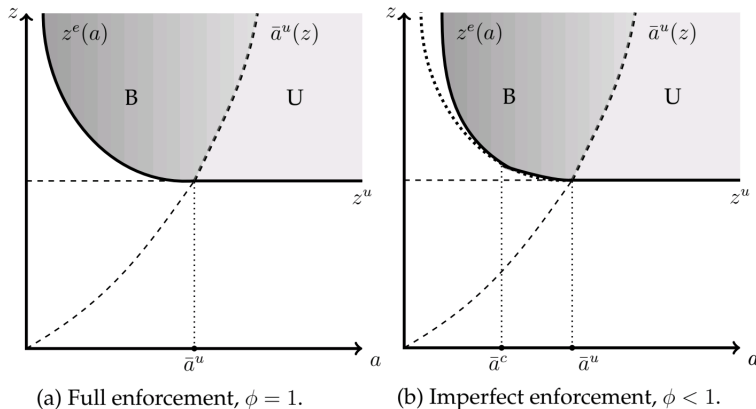
Current income flow: $\mathcal{I}(a, z) = \max\{w, \pi^*(a, z)\}$

Occupational Choice:

$$o(a, z) = \begin{cases} 1, & \text{if being entrepreneur} \\ 0, & \text{if being worker} \end{cases}$$

Model: Static Optimization

Figure 1: Occupational choice, $\chi = 0$.



Notes: The light gray shaded area, **U**, contains the measure of unconstrained entrepreneurs. The dark gray shaded area, **B**, displays the measure of constrained borrowers. The white area below the curve $z^e(a)$ represents the measure of workers.

Model: Dynamic Optimization

Budget constraint: $\dot{a}(a, z) = \mathcal{I}(a, z) + (r - \delta)a - c(a, z)$

HJB: $\rho V(a, z) =$

$$\max_c u(c) + \partial_a V(a, z)(\mathcal{I}(a, z) + (r - \delta)a - c) + \gamma \left[\int_{\mathbb{Z}} V(a, \tilde{z}) \mu(\tilde{z}) d\tilde{z} - V(a, z) \right]$$

Model: Equilibrium

Stationary competitive equilibrium:

- Stationary distribution $H(a, z)$
- prices: $w, r, \tilde{r}(a, z)$
- $k(a, z), n(a, z), o(a, z), c(a, z)$
- $V(a, z)$

s.t.

- Given the prices, the policy functions solve the recursive problem with the financial contracts

- Capital: $K := \int_{\{o(a,z)=1\}} k(a, z) H(da, dz) = \int a H(da, dz)$

- Labor: $N := \int_{\{o(a,z)=1\}} n(a, z) H(da, dz) = \int_{\{o(a,z)=0\}} H(da, dz)$

- Final goods: $\int c(a, z) H(da, dz) + \int_{\{o(a,z)=1, k(a,z)>a\}} \tau(a, z) H(da, dz) = \int_{\{o(a,z)=1\}} y(a, z) H(da, dz) - \delta K$

- Kolmogorov forward equation:

$$0 = -\frac{d}{da} [\dot{a}(a, z) h(a, z)] - \gamma h(a, z) + \gamma \mu(z) \int h(a, \tilde{z}) d\tilde{z}$$

Table 3: Calibration and Model Fit

Parameter Values		
Parameter	Description	Value
<i>2 assigned parameters</i>		
σ	Coefficient of relative risk aversion	1.50
δ	Depreciation rate	0.06
<i>11 calibrated parameters</i>		
ρ	Subjective discount rate	0.25
α	Elast. of y with respect to k	0.29
θ	Elast. of y with respect to n	0.35
κ	Fixed cost of production	1.02
η	Curvature of the Pareto distr.	3.47
γ	New productivity arrival rate	0.15
ϕ	Enforcement parameter	0.12
τ_0	Interm. costs - independent factor	0.01
τ_a	Interm. costs - elast. of assets	0.68
τ_z	Interm. costs - elast. of productivity	0.23
χ	Bank barg. power in a loan	0.03
Model Fit		
12 Targeted Moments		
	Data	Model
Risk-free bond rate	0.02	0.02
Capital's share of income	0.40	0.28
Investment rate	0.17	0.11
Top 10% earners' income share	0.56	0.59
Average firm size	21.0	21.2
Top 10% firms' employment share	0.78	0.68
Firm exit rate	0.08	0.05
External finance to GDP ratio	0.49	0.52
Fraction firms with credit	0.35	0.40
Average spread (unweighted)	0.38	0.40
Average spread (credit-weighted)	0.06	0.06
Standard deviation of spread	0.32	0.33

Table 4: Impacts of Credit Frictions on Development

Value Relative to Perfect Credit:	Perfect Credit (1)	Benchmark (2)	No Quant. Constr. (3)	No Spread Frictions (4)	No Spread Disp. (5)	All Quant. Constr. (6)
<i>Aggregate values relative to perfect credit world:</i>						
GDP	1.00	0.69	0.76	0.90	0.80	0.86
TFP	1.00	0.75	0.82	0.95	0.88	0.92
Wage	1.00	0.65	0.76	0.92	0.84	0.87
Capital	1.00	0.74	0.78	0.84	0.73	0.80
Credit/GDP	1.00	0.47	0.72	0.66	0.47	0.48
Interest rate	0.13	0.02	0.05	0.06	0.01	0.01
Firm growth	0.01	0.06	0.05	0.03	0.03	0.03
Exit rate	0.12	0.05	0.12	0.12	0.12	0.12
Avg. firm size	9	21	19	9	10	8

Column (2) parameter values are those calibrated in Table 3. Relative to these values, Column (1):

$\tau_i = \chi = 0$ and $\phi = 1$, Column (3): $\phi = 1$, Column (4): $\chi = \tau_i = 0$, Column (5): $\tau_a = \tau_z = \chi = 0$ and $\tau_0 = 0.09$, and Column (6): $\tau_i = \chi = 0$ and $\phi = 0.05$, calibrated to match credit/GDP in the benchmark and data.

- Credit spreads are the dominant source of friction in the benchmark model.
- Counterfactuals: aggregate impacts from uniform spread or from quantity constraints along are mild.

Quantitative Analysis

Table 5: Isolated Impacts of Spread-Causing Frictions

	Eliminating Frictions				Single Friction Calibrations				
		No Market Power	No Uniform Cost	No a -depend. Cost	No z -depend. Cost	All Market Power [†]	All Uniform Cost	All a -depend. Cost	All z -depend. Cost
	Benchmk (1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
<i>Aggregate values relative to perfect credit world:</i>									
GDP	0.69	0.69	0.69	0.75	0.72	0.55	0.82	0.40	0.82
TFP	0.75	0.75	0.76	0.80	0.77	0.64	0.90	0.52	0.88
Wage	0.65	0.66	0.66	0.72	0.67	0.51	0.87	0.47	0.83
Capital	0.74	0.73	0.75	0.81	0.78	0.60	0.75	0.43	0.80
Credit/GDP	0.47	0.47	0.49	0.48	0.57	0.66	0.53	0.11	0.54
<i>Firm credit spread moments:</i>									
Interest rate	0.02	0.02	0.02	0.03	0.04	0.04	0.03	0.00	0.02
Avg. (weighted)	0.06	0.06	0.05	0.05	0.03	0.02	0.06	0.06	0.06
Avg. (unweighted)	0.40	0.48	0.39	0.08	0.29	0.02	0.06	22.9	0.11
Std. deviation	0.33	0.33	0.33	0.02	0.31	0.01	0.00	18.4	0.02
Frac. with credit	0.40	0.47	0.41	0.36	0.48	0.84	0.93	0.01	0.81
Firm growth	0.06	0.06	0.07	0.01	0.08	0.04	0.03	0.02	0.03
Exit rate	0.05	0.06	0.06	0.04	0.06	0.01	0.12	0.00	0.12
Avg. firm size	21	19	21	20	21	23	8	7	12

Column (1) parameter values are those calibrated in Table 3. Relative to these values, Column (2): $\chi = 0$, Column (3): $\tau_0 = 0$, Column (4): $\tau_a = 0$, Column (5): $\tau_z = 0$, Column (6): $\tau_1 = 0$ and $\chi = 0.57$, Column (7): $\tau_a = \tau_z = \chi = 0$ and $\tau_0 = 0.06$, Column (8): $\tau_0 = \tau_z = \chi = 0$ and $\tau_a = 37.7$, and Column (9): $\tau_0 = \tau_a = \chi = 0$ and $\tau_z = 0.30$. For Columns (7)-(9), the calibrated value is chosen to match the weighted spread in the benchmark.

[†]: It is difficult for the code to converge when market power is very high. Hence, Column (6) generates a spread of around 2% by quadrupling the market power relative to the benchmark.

- τ_a plays the largest independent role

Life Cycle Dynamics in Data and Model: Spread Patterns

Table 6: Spreads and Firm Characteristics - Data vs. Model

	Data [†]			Model		
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	0.0067*** [57.04]	0.0416*** [12.62]	0.0307*** [5.37]	0.4419*** [1974.86]	0.1550*** [585.91]	0.0924*** [388.92]
Age	-0.0007*** [-66.92]	-0.0041*** [-12.31]	-0.0042*** [-13.24]	-0.0240*** [-1443.73]	-0.0078*** [-569.91]	-0.0056*** [-458.34]
Age ²	0.0000*** [57.59]	0.0000*** [11.63]	0.0000*** [12.50]	0.0003*** [958.16]	0.0001*** [580.37]	0.0001*** [957.74]
Wagebill (ln)			0.0015*** [2.75]			0.0303*** [1406.18]
Loan controls	Yes	Yes	Yes	-	-	-
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	No	Yes	Yes	No	Yes	Yes
Observations	11,846,236	11,846,236	10,850,982	7,407,211	7,407,211	7,407,211
R-squared	0.000	0.539	0.541	0.531	0.931	0.947
RMSE	0.226	0.179	0.178	0.054	0.021	0.019

The dependent variable is $\ln(1 + spread)_{i,t}$, residualized of loan controls, time FEs in the data.

[†] The "Data" results reproduce the residual regressions in Columns (2), (3), and (5) of Table 2.

t statistics in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

- Both model and data show a significant, declining but convex life cycle of spreads.
- Controlling for fixed effects, the spread is increasing in wage bill in both model and data.

Life Cycle Dynamics in Data and Model: Firm Growth

$$\frac{n_{it}}{n_{it-1}} = \beta_i + \beta_a h_{it} + \beta_{a^2} h_{it}^2 + \beta_{\text{loan}} \mathbb{I}_{\text{loan}_{it}} + \beta_{\tilde{r}} \ln(1 + \text{spread})_{it} + \epsilon_{it}$$

Table 7: Firm Growth Patterns $\log(\frac{n_{it}}{n_{it-1}})$ - Data vs. Model

	Data			Model			
	(1)	(2)	(3)	(4)	(5)	(6)	(7) [†]
Constant	0.0647** [2.23]	0.2623*** [5.82]	0.2456*** [5.42]	0.0450*** [41.47]	0.1555*** [47.20]	-0.1693*** [-53.91]	0.1356*** [41.18]
Age	-0.0057*** [-22.00]	-0.0132*** [-32.39]	-0.0134*** [-32.85]	-0.0049*** [-61.50]	-0.0088*** [-52.80]	0.0067*** [42.16]	-0.0084*** [-50.01]
Age ²	0.0000*** [21.12]	0.0001*** [31.49]	0.0001*** [31.97]	0.0001*** [66.53]	0.0001*** [54.82]	-0.0001*** [-42.91]	0.0001*** [52.05]
Have loan			0.0310*** [15.87]			0.2834*** [143.33]	0.0207*** [114.81]
ln(1+spread)			0.0415*** [9.40]			1.9889*** [115.91]	0.0490*** [86.97]
Loan controls	Yes	Yes	Yes	-	-	-	-
Time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	No	Yes	Yes	No	Yes	Yes	Yes
Observations	9,167,466	9,167,466	9,127,081	6,992,261	6,992,261	6,992,261	6,992,261
R-squared	0.049	0.286	0.287	0.0008	0.0299	0.1312	0.0330
RMSE	0.525	0.527	0.526	0.226	0.230	0.218	0.230

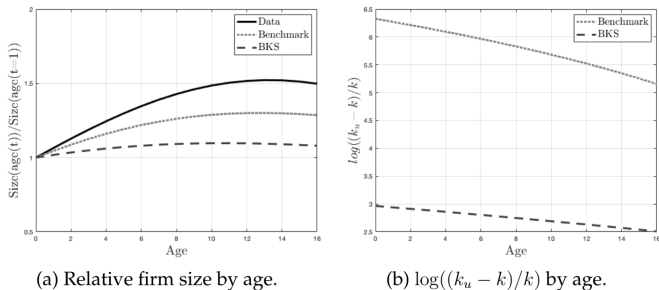
[†] In the simulated results in Column (7), random normal, i.i.d. errors have been added to the spread and got loan controls, calibrated to match the unexplained spread variation in the model to the data.

t statistics in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

- Concave increase in size over the first 15 years

Life Cycle Dynamics in Data and Model: Firm Growth

Figure 2: Firm Dynamics



Notes: Benchmark is the full model with both a quantity constraint and spread frictions. BKS is a model with only a quantity constraint (as in Buera, Kaboski, and Shin, 2011) recalibrated to match the same level of credit/GDP as in Column (6) of Table 4.

- Aggregate misallocation is linked tightly to faster firm growth: larger initial misallocation and incentive to grow quickly

Conclusion

- Using a comprehensive firm-level credit registry, the paper documented both high levels and high dispersion in credit spreads to Brazilian firms.
- Developed a quantitative dynamic general equilibrium model in which dispersion in spreads arises from intermediation costs and market power.
- Calibrating to the Brazilian data, showed that, for equivalent levels of external financing, dispersion has more profound impacts on aggregate development than single-price credit rationing and yields firm dynamics that are more consistent with observed patterns
- Causes of credit spreads: Banking sector?
- Margins affecting aggregates?

$$\max_{\bar{k} \geq k \geq a, \tilde{r}} ((\tilde{r} - r - \tau(a, z)(k - a)))^\chi (\pi(k, \tilde{r}; a, z) - \tilde{w}(a, z))^{1-\chi}$$

Equation (6):

$$\tilde{r}(a, z) = \chi \frac{\pi^b(a, z) - \tilde{w}(a, z)}{k^b - a} + (1 - \chi)(r + \tau(a, z))$$

where

$$\begin{aligned}\pi^b(a, z) &= y^b(a, z) - wn^b(a, z) - ra - \kappa \\ z(k^b(a, z))^\alpha \theta(n^b(a, z))^{\theta-1} &= w\end{aligned}$$

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